

HARSHA DURAIRAJ

Aspiring Robotics & AI Engineer

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 Uppuviduthi, Thiruchitrambalam, pattukkottai, Thanjavur , Tamil Nadu - 614628 ,India

PROFILE

I am a 3rd-year Robotics and Artificial Intelligence student passionate about developing application-oriented systems that integrate hardware and software. My interests include creating innovative technologies in robotics, AI, healthcare, and agriculture to solve real-world problems and support societal development.

EDUCATION

SASTRA Deemed University, Thanjavur.

B.Tech Robotics And AI

2023 - 2027

CGPA:7.758 (up to 5th semester)

Muvendar Matric Higher Secondary School, Peravurani.

Class-12 th

2023

89.66%

Muvendar Matric Higher Secondary School, Peravurani.

Class-10 th

2021

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SKILLS

Programming & Development:

Python, MATLAB, HTML, Machine Learning, Prompt Engineering.

Tools:

Arduino IDE, ROS2, Visual Studio, Colab.

Professional Attributes:

Leadership, Team Coordination, Adaptability, Time Management.

PROJECTS

FARM RUN – AI-Powered Grid Adventure Game Using A* Pathfinding

Designed and developed an 8×8 grid-based strategic adventure game where the player navigates toward a home location while avoiding traps and AI-controlled enemies, implementing intelligent pathfinding using the A* algorithm.

Autonomous Room-Level Delivery Robot Using MCL

Developed an autonomous indoor delivery robot implementing Monte Carlo Localization (MCL) for precise room-level navigation and obstacle avoidance.

Adaptive Exoskeleton with EEG-EMG-Based Fatigue Detection

Developed a machine learning pipeline integrating EEG, EMG, and text embeddings to estimate human fatigue levels in real time. Implemented a fusion model combining deep CNN and logistic regression outputs to generate a unified fatigue score (0–100) that dynamically adjusts exoskeleton support for enhanced safety and efficiency.

CERTIFICATES

- **Python Programming Certification** – GUVI
- **AI for India 2.0 Course** – GUVI
- **Introduction to Industrial and Mobile Robots** – MHI
- **MATLAB Onramp Course** – MathWorks
- **Fundamentals and Basics of Robotics Workshop** – Robotics Club @ SASTRA (RCS)

LANGUAGES : Tamil (Native Proficiency) , English (Full Professional Proficiency)